

## Skill Check 2: Exploring Data

### Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human. Cell phones are to be handed in to the instructor and all other digital devices must be stored in bags underneath your desk or against the wall.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

**Printed Name:** \_\_\_\_\_

**Section:** \_\_\_\_\_

**Signed:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Include this signed page as the first page of your submitted work.**

**Skill Check Instructions**

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

## Question 1: Loading, Cleaning, and Analyzing Data

You have a data file that you'd like to analyze in R named `toucan_beaks.csv` that contains a bird ID number (`number`), measurements of beak length (`beak_length`), depth (`beak_depth`), and the field site (`site`). The first two lines of this file contain information on the dates and times toucans were captured and measured but do not contain the beak measurements that you wish to analyze.

- Write the code to load the data set into R as a data frame named `bird_beaks`.
- Calculate the mean lengths of beaks for each field site (they are `costa_rica` and `venezuela`).
- What types of data are each of these columns (except bird ID number)? (Note: asking for general data type relevant to analysis, not the data class.)
- If you wanted to plot beak length versus file site, what would be the best type of plot for visualizing these data?

**Question 2: Plotting Basics**

Using the `penguins` data frame, make a scatter plot of bill length versus bill depth and color the points by species.

|                |                               |                                                                                     |
|----------------|-------------------------------|-------------------------------------------------------------------------------------|
| ▼ penguins     | 344 obs. of 8 variables       |  |
| \$ species     | : Factor w/ 3 levels "Ade..." |                                                                                     |
| \$ island      | : Factor w/ 3 levels "Bis..." |                                                                                     |
| \$ bill_len    | : num 39.1 39.5 40.3 NA ...   |                                                                                     |
| \$ bill_dep    | : num 18.7 17.4 18 NA 19...   |                                                                                     |
| \$ flipper_len | : int 181 186 195 NA 193...   |                                                                                     |
| \$ body_mass   | : int 3750 3800 3250 NA ...   |                                                                                     |
| \$ sex         | : Factor w/ 2 levels "fem..." |                                                                                     |
| \$ year        | : int 2007 2007 2007 200...   |                                                                                     |

a. With base graphics.

b. With `ggplot2` package.

**Question 3: Visualizing and Communicating Data**

Describe the ethical obligation you have when visualizing data. Why is this important?